

ER RELATIONAL SCHEMA ATTRIBUTES

SECTION 1 — PEOPLE

sql

```
CREATE TABLE "User" (  
  user_id SERIAL PRIMARY KEY,  
  full_name VARCHAR(100) NOT NULL,  
  email VARCHAR(120) UNIQUE NOT NULL,  
  phone VARCHAR(20) UNIQUE NOT NULL,  
  date_of_birth DATE NOT NULL,  
  home_city VARCHAR(100),  
  account_status VARCHAR(20) DEFAULT 'ACTIVE'  
  CHECK (account_status IN  
    ('ACTIVE', 'SUSPENDED', 'DELETED')),  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

sql

```
CREATE TABLE Vendor (  
  vendor_id SERIAL PRIMARY KEY,  
  company_name VARCHAR(150) NOT NULL,  
  vendor_type VARCHAR(20) NOT NULL  
  CHECK (vendor_type IN  
    ('TRANSPORT', 'HOTEL', 'PACKAGE')),  
  gst_number VARCHAR(50) UNIQUE NOT NULL,  
  registered_address TEXT NOT NULL,  
  website VARCHAR(150),  
  contact_name VARCHAR(100) NOT NULL,  
  contact_phone VARCHAR(20),  
  contact_email VARCHAR(120),  
  is_active BOOLEAN DEFAULT TRUE,  
  created_at TIMESTAMP DEFAULT NOW()  
);
```

SECTION 2 — TRANSPORT SUPPLY SIDE

sql

```
CREATE TABLE TransportRoute (  
  route_id SERIAL PRIMARY KEY,  
  transport_mode VARCHAR(20) NOT NULL  
  CHECK (transport_mode IN
```

```

                                ('FLIGHT', 'TRAIN', 'BUS')),
    origin_city      VARCHAR(100) NOT NULL,
    destination_city VARCHAR(100) NOT NULL,
    vehicle_number   VARCHAR(50)
);

```

Why no vendor_id here? Multiple vendors operate the same route. IndiGo AND Air India both fly Mumbai → Delhi. vendor_id lives on RouteSchedule instead — each vendor defines their own schedule on a shared route.

sql

```

CREATE TABLE RouteSchedule (
    schedule_id SERIAL PRIMARY KEY,
    route_id    INT NOT NULL
                REFERENCES TransportRoute(route_id)
                ON DELETE CASCADE,
    vendor_id   INT NOT NULL
                REFERENCES Vendor(vendor_id),
    departure_time TIME NOT NULL,
    arrival_time  TIME NOT NULL,
    is_active    BOOLEAN DEFAULT TRUE
);

```

Why no days_of_week here? Days of week is a multi-valued attribute — one schedule can run on multiple days. Storing "MON,WED,FRI" as a string violates First Normal Form. It gets its own table below.

sql

- Multi-valued attribute: days_of_week split into own table

```

CREATE TABLE
ScheduleOperatingDay ( schedule_id INT NOT NULL REFERENCES RouteSchedule(schedule_id) ON
DELETE CASCADE, day_of_week VARCHAR(10) NOT NULL CHECK (day_of_week IN
('MON','TUE','WED','THU','FRI','SAT','SUN')), PRIMARY KEY (schedule_id, day_of_week));

```

sql

- Weak entity – depends on RouteSchedule
- ```

CREATE TABLE ScheduleClass (class_id SERIAL PRIMARY KEY, schedule_id INT NOT NULL
REFERENCES RouteSchedule(schedule_id) ON DELETE CASCADE, class_name VARCHAR(50) NOT NULL
CHECK (class_name IN ('ECONOMY','BUSINESS','FIRST','SLEEPER','3AC','2AC','1AC')),
total_seats INT NOT NULL CHECK (total_seats > 0), base_fare DECIMAL(10,2) NOT NULL CHECK
(base_fare > 0));

```

sql

- Weak entity – depends on RouteSchedule + ScheduleClass- One row = one real bookable slot (schedule × class × date)  

```

CREATE TABLE ScheduleDeparture (departure_id SERIAL

```

```
PRIMARY KEY, schedule_id INT NOT NULL REFERENCES RouteSchedule(schedule_id) ON DELETE CASCADE, class_id INT NOT NULL REFERENCES ScheduleClass(class_id) ON DELETE CASCADE, departure_date DATE NOT NULL, available_seats INT NOT NULL CHECK (available_seats >= 0), dynamic_fare DECIMAL(10,2) NOT NULL CHECK (dynamic_fare > 0), status VARCHAR(20) DEFAULT 'SCHEDULED' CHECK (status IN ('SCHEDULED','CANCELLED','COMPLETED')), UNIQUE (schedule_id, class_id, departure_date));
```

**Why UNIQUE constraint on (schedule\_id, class\_id, departure\_date)?** Because one schedule + one class + one date can only have one departure slot. Prevents duplicate rows being auto-generated by the system.

## SECTION 3 — HOTEL SUPPLY SIDE

sql

```
CREATE TABLE Hotel (
 hotel_id SERIAL PRIMARY KEY,
 vendor_id INT NOT NULL
 REFERENCES Vendor(vendor_id),
 hotel_name VARCHAR(150) NOT NULL,
 city VARCHAR(100) NOT NULL,
 full_address TEXT,
 star_rating INT CHECK (star_rating BETWEEN 1 AND 5),
 checkin_time TIME,
 checkout_time TIME,
 is_active BOOLEAN DEFAULT TRUE
);
```

**Why no amenities column here?** Amenities is a multi-valued attribute — a hotel has many amenities (Pool, Gym, Spa, WiFi, Parking). Storing as TEXT violates 1NF. Gets its own table below.

sql

- Multi-valued attribute: amenities split into own table
- CREATE TABLE HotelAmenity ( hotel\_id INT NOT NULL REFERENCES Hotel(hotel\_id) ON DELETE CASCADE, amenity VARCHAR(100) NOT NULL, PRIMARY KEY (hotel\_id, amenity));

sql

- Weak entity – depends on Hotel
- CREATE TABLE RoomCategory ( room\_category\_id SERIAL PRIMARY KEY, hotel\_id INT NOT NULL REFERENCES Hotel(hotel\_id) ON DELETE CASCADE, category\_name VARCHAR(100) NOT NULL, max\_guests INT NOT NULL CHECK (max\_guests > 0), price\_per\_night DECIMAL(10,2) NOT NULL CHECK (price\_per\_night > 0), total\_rooms INT NOT NULL CHECK (total\_rooms > 0));

sql

- - Weak entity – depends on RoomCategory- Composite PK: one row per (room type × date)
- ```
CREATE TABLE RoomAvailability ( room_category_id INT NOT NULL REFERENCES RoomCategory(room_category_id) ON DELETE CASCADE, available_date DATE NOT NULL, available_rooms INT NOT NULL CHECK (available_rooms >= 0), PRIMARY KEY (room_category_id, available_date));
```

SECTION 4 — PACKAGE SUPPLY SIDE

sql

```
CREATE TABLE HolidayPackage (  
  package_id SERIAL PRIMARY KEY,  
  vendor_id INT NOT NULL  
  REFERENCES Vendor(vendor_id),  
  package_name VARCHAR(150) NOT NULL,  
  theme VARCHAR(100)  
  CHECK (theme IN  
  ('BEACH', 'ADVENTURE', 'HONEYMOON',  
  'FAMILY', 'PILGRIMAGE', 'CULTURAL',  
  'WILDLIFE')),  
  duration_days INT NOT NULL CHECK (duration_days > 0),  
  duration_nights INT NOT NULL CHECK (duration_nights > 0),  
  base_price_per_person DECIMAL(10,2) NOT NULL,  
  is_active BOOLEAN DEFAULT TRUE  
);
```

Why no destinations or inclusions columns? Both are multi-valued. A package visits multiple destinations and includes multiple things (Flight + Hotel + Meals + Transfers). Both get their own tables below.

sql

- - Multi-valued attribute: destinations
- ```
CREATE TABLE PackageDestination (package_id INT NOT NULL REFERENCES HolidayPackage(package_id) ON DELETE CASCADE, city VARCHAR(100) NOT NULL, visit_order INT, PRIMARY KEY (package_id, city));
```

**visit\_order** stores the sequence of cities visited. "Day 1 Mumbai → Day 3 Goa → Day 6 Kerala" — the order matters for itinerary display.

sql

- - Multi-valued attribute: inclusions

- ```
CREATE TABLE PackageInclusion ( package_id INT NOT NULL REFERENCES HolidayPackage(package_id) ON DELETE CASCADE, inclusion VARCHAR(100) NOT NULL CHECK (inclusion IN ('FLIGHT','HOTEL','BREAKFAST','ALL_MEALS','TRANSFERS','SIGHTSEEING','VISA')), PRIMARY KEY (package_id, inclusion));
```

sql

- - Weak entity – depends on HolidayPackage
- ```
CREATE TABLE PackageDeparture (package_departure_id SERIAL PRIMARY KEY, package_id INT NOT NULL REFERENCES HolidayPackage(package_id) ON DELETE CASCADE, departure_date DATE NOT NULL, total_seats INT NOT NULL CHECK (total_seats > 0), available_seats INT NOT NULL CHECK (available_seats >= 0), price_per_person DECIMAL(10,2) NOT NULL, status VARCHAR(20) DEFAULT 'OPEN' CHECK (status IN ('OPEN','FULL','CANCELLED')), UNIQUE (package_id, departure_date));
```

## SECTION 5 — PROMO CODES

sql

```
CREATE TABLE PromoCode (
 promo_id SERIAL PRIMARY KEY,
 code VARCHAR(50) UNIQUE NOT NULL,
 discount_type VARCHAR(20) NOT NULL
 CHECK (discount_type IN
 ('FLAT','PERCENTAGE')),
 discount_value DECIMAL(10,2) NOT NULL CHECK (discount_value > 0),
 max_discount_cap DECIMAL(10,2),
 min_booking_amount DECIMAL(10,2) DEFAULT 0,
 valid_from DATE NOT NULL,
 valid_to DATE NOT NULL,
 applicable_on VARCHAR(20) DEFAULT 'ALL'
 CHECK (applicable_on IN
 ('TRANSPORT','HOTEL',
 'PACKAGE','ALL')),
 max_uses INT,
 used_count INT DEFAULT 0,
 is_active BOOLEAN DEFAULT TRUE,
 CHECK (valid_to > valid_from)
);
```

**Why PromoCode before Booking?** Booking has a FK to PromoCode (promo\_id). The referenced table must exist first.

## SECTION 6 — BOOKING LAYER

## Superclass first, then subclasses

sql

- - SUPERCLASS – every booking regardless of type  

```
CREATE TABLE Booking (booking_id SERIAL PRIMARY KEY, user_id INT NOT NULL REFERENCES "User"(user_id), booking_type VARCHAR(20) NOT NULL CHECK (booking_type IN ('TRANSPORT','HOTEL','PACKAGE')), booked_at TIMESTAMP DEFAULT NOW(), status VARCHAR(20) DEFAULT 'CONFIRMED' CHECK (status IN ('CONFIRMED','CANCELLED','COMPLETED')), base_amount DECIMAL(10,2) NOT NULL CHECK (base_amount > 0), discount_amount DECIMAL(10,2) DEFAULT 0, final_amount DECIMAL(10,2) NOT NULL CHECK (final_amount >= 0), promo_id INT REFERENCES PromoCode(promo_id), payment_method VARCHAR(20) CHECK (payment_method IN ('UPI','CARD','NETBANKING','WALLET')), payment_ref VARCHAR(100), CHECK (final_amount = base_amount - discount_amount));
```

sql

- - SUBCLASS 1 – result of Aggregation Box 1- (User selects ScheduleDeparture → confirms → TransportBooking)  

```
CREATE TABLE TransportBooking (booking_id INT PRIMARY KEY REFERENCES Booking(booking_id) ON DELETE CASCADE, departure_id INT NOT NULL REFERENCES ScheduleDeparture(departure_id), num_seats INT NOT NULL CHECK (num_seats > 0));
```

sql

- - SUBCLASS 2 – result of Aggregation Box 2- (User browses RoomCategory → confirms → HotelBooking)  

```
CREATE TABLE HotelBooking (booking_id INT PRIMARY KEY REFERENCES Booking(booking_id) ON DELETE CASCADE, hotel_id INT NOT NULL REFERENCES Hotel(hotel_id), room_category_id INT NOT NULL REFERENCES RoomCategory(room_category_id), checkin_date DATE NOT NULL, checkout_date DATE NOT NULL, num_rooms INT NOT NULL CHECK (num_rooms > 0), num_guests INT NOT NULL CHECK (num_guests > 0), CHECK (checkout_date > checkin_date));
```

sql

- - SUBCLASS 3 – result of Aggregation Box 3- (User explores PackageDeparture → confirms → PackageBooking)  

```
CREATE TABLE PackageBooking (booking_id INT PRIMARY KEY REFERENCES Booking(booking_id) ON DELETE CASCADE, package_departure_id INT NOT NULL REFERENCES PackageDeparture(package_departure_id), num_travellers INT NOT NULL CHECK (num_travellers > 0));
```

---

## SECTION 7 — PEOPLE ON BOOKINGS

### Weak entities under each subclass

sql

- - Weak entity under TransportBooking
- ```
CREATE TABLE Passenger ( passenger_id SERIAL PRIMARY KEY, booking_id INT NOT NULL REFERENCES TransportBooking(booking_id) ON DELETE CASCADE, full_name VARCHAR(100) NOT
```

```
NULL, age INT NOT NULL CHECK (age > 0), gender VARCHAR(10) CHECK (gender IN ('MALE','FEMALE','OTHER')), govt_id_type VARCHAR(20) CHECK (govt_id_type IN ('AADHAR','PASSPORT','PAN')), govt_id_number VARCHAR(50) NOT NULL, seat_number VARCHAR(10));
```

sql

- - Weak entity under HotelBooking
- ```
CREATE TABLE HotelGuest (guest_id SERIAL PRIMARY KEY, booking_id INT NOT NULL REFERENCES HotelBooking(booking_id) ON DELETE CASCADE, guest_name VARCHAR(100) NOT NULL);
```

sql

- - Weak entity under PackageBooking
- ```
CREATE TABLE PackageTraveller ( package_traveller_id SERIAL PRIMARY KEY, booking_id INT NOT NULL REFERENCES PackageBooking(booking_id) ON DELETE CASCADE, full_name VARCHAR(100) NOT NULL, age INT NOT NULL CHECK (age > 0), gender VARCHAR(10) CHECK (gender IN ('MALE','FEMALE','OTHER')), govt_id_type VARCHAR(20) CHECK (govt_id_type IN ('AADHAR','PASSPORT','PAN')), govt_id_number VARCHAR(50) NOT NULL);
```

SECTION 8 — POST BOOKING FEATURES

sql

- - 1:1 with Booking – insurance on a booking
- ```
CREATE TABLE Insurance (insurance_id SERIAL PRIMARY KEY, booking_id INT UNIQUE NOT NULL REFERENCES Booking(booking_id) ON DELETE CASCADE, provider_name VARCHAR(150) NOT NULL, plan_name VARCHAR(150) NOT NULL, plan_type VARCHAR(20) CHECK (plan_type IN ('SINGLE_TRIP','MULTI_TRIP')), coverage_summary TEXT, premium_amount DECIMAL(10,2) NOT NULL CHECK (premium_amount > 0), valid_from DATE NOT NULL, valid_to DATE NOT NULL, claim_status VARCHAR(20) DEFAULT 'NOT_CLAIMED' CHECK (claim_status IN ('NOT_CLAIMED','CLAIM_FILED','SETTLED')), CHECK (valid_to > valid_from));
```

sql

- - 1:1 with Booking – one cancellation per booking
- ```
CREATE TABLE CancellationRefund ( cancellation_id SERIAL PRIMARY KEY, booking_id INT UNIQUE NOT NULL REFERENCES Booking(booking_id), cancelled_at TIMESTAMP DEFAULT NOW(), reason TEXT, refund_amount DECIMAL(10,2) CHECK (refund_amount >= 0), refund_status VARCHAR(20) DEFAULT 'PENDING' CHECK (refund_status IN ('PENDING','PROCESSED','REJECTED')), processed_at DATE);
```

sql

- - 1:1 with Booking – one review per booking

-

- ```
CREATE TABLE Review (review_id SERIAL PRIMARY KEY, booking_id INT UNIQUE NOT NULL
REFERENCES Booking(booking_id), user_id INT NOT NULL REFERENCES "User"(user_id),
entity_type VARCHAR(20) NOT NULL CHECK (entity_type IN ('TRANSPORT','HOTEL','PACKAGE')),
entity_id INT NOT NULL, star_rating INT NOT NULL CHECK (star_rating BETWEEN 1 AND 5),
title VARCHAR(150), comment TEXT, reviewed_at TIMESTAMP DEFAULT NOW());
```

## sql

```
- Standalone computed cache – no FK constraint- Populated by background job onlyCREATE TABLE
DemandTag (tag_id SERIAL PRIMARY KEY, service_type VARCHAR(20) NOT NULL CHECK (service_type
IN ('TRANSPORT','HOTEL','PACKAGE')), service_id INT NOT NULL, tag_value VARCHAR(30) NOT NULL
CHECK (tag_value IN ('HIGH_DEMAND','FILLING_FAST', 'LOW_AVAILABILITY')), last_computed
TIMESTAMP DEFAULT NOW());````--
COMPLETE TABLE COUNT
```PEOPLE (2)-----
```

```
1. User2. Vendor
TRANSPORT SUPPLY (5)-----
3. TransportRoute
4. RouteSchedule
5. ScheduleOperatingDay ← from multivalued days_of_week
6. ScheduleClass
7. ScheduleDeparture
```

```
HOTEL SUPPLY (4)-----
8. Hotel
9. HotelAmenity ← from multivalued amenities
10. RoomCategory
11. RoomAvailability
PACKAGE SUPPLY (4)-----
12. HolidayPackage
13. PackageDestination ← from multivalued destinations
14. PackageInclusion ← from multivalued inclusions
15. PackageDeparture
```

```
PROMO (1)-----
16. PromoCode
```

```
BOOKING LAYER (6)-----
17. Booking ← superclass
18. TransportBooking ← subclass
19. HotelBooking ← subclass
20. PackageBooking ← subclass
21. Passenger ← weak under TransportBooking
22. HotelGuest ← weak under HotelBooking
23. PackageTraveller ← weak under PackageBooking
```

POST BOOKING FEATURES (4)

24. Insurance

25. CancellationRefund

26. Review

27. DemandTag ← standalone cache

TOTAL: 27 TABLES4 born from multivalued attributes

(ScheduleOperatingDay, HotelAmenity, PackageDestination, PackageInclusion)23 core relational tables